

How do you define
Soft?



MicroTouch
from Micro Powders



MICRO POWDERS, INC.



MicroTouch

Soft Touch Coating Additives



INTRODUCTION

Coating formulators now have the ability to modify the tactile surface properties of their products with MicroTouch. Based on aliphatic polyurethane technology, MicroTouch additives are micron-sized elastic particles that distort when disturbed by a physical force (such as finger touch or abrasive action) and then return to their original spherical shape when the force is removed. These products can be used in combination with many different types of coating resins to achieve effects that range from satiny smooth to leathery to rubbery. MicroTouch additives are highly resilient particles that can be used to modify and enhance the feel of a coating surface. These additives will provide a matte appearance as well as superior scratch, burnish, and mar resistance. MicroTouch additives are available in several particle sizes, each of which brings different tactile properties. MicroTouch additives are an ideal alternative to formaldehyde-based materials such as polyureas.

MicroTouch additives will improve chemical resistance and hydrolytic stability. Taber abrasion resistance can be dramatically improved with an addition of 2-3% of MicroTouch.

MicroTouch additives can be used to modify coatings designed for plastics, metals, paper, wood, leather, and other textiles. Common industrial applications include electronics (computers, laptops, mobile phones), automotive interiors, wall coverings, parquet and vinyl flooring, inks, and furniture.

FORMULATION NOTES

It is important to understand that the ultimate “feel” of a coating formulated with MicroTouch is highly dependent on the choice of resin system. Harder resin systems formulated with MicroTouch will provide a surface with a smooth, satiny feel. Softer resin systems can provide surfaces that have a rubbery, anti-slip feel. Lower percentages (2-6%) of MicroTouch will provide gloss reduction and

abrasion resistance with limited tactile effects. At higher percentages (6-10%), more pronounced tactile properties can be achieved.

The formulated examples presented in this brochure demonstrate some of the tactile effects that can be achieved by combining MicroTouch additives with different types of coating resins. These examples should not be considered as fully formulated commercial coatings. They are presented strictly to show the range of soft touch effects that can be achieved.

PROCESSING NOTES

MicroTouch additives are virtually indestructible and are easy to incorporate into coatings with an impeller disk disperser such as a Conn ITT blade (or similar). A tip speed of around 3-3.5 m/s should be sufficient to wet and disperse the product into a coating. In solvent based systems, additional shear energy may be required to disperse and deagglomerate the product.

HIGH TEMPERATURE AND SOLVENT BASED APPLICATIONS

MicroTouch additives are suitable for applications with curing temperatures up to 100-120 °C. For applications in the 120-140 °C range, the products may show some yellowing. Applications exceeding 140 °C are not recommended as the particles will swell and/or stick together.

MicroTouch additives will swell and absorb solvent if slurried directly into alcohol (ethanol, isopropanol) or other highly polar solvents. They should instead be dispersed into the alcohol based resin system to disperse and stabilize the particles.

MicroTouch additives should not be used in coating systems based on dimethylformamide (DMF) as the particles will swell significantly.

Product Guide

Typical Properties	MicroTouch 800F	MicroTouch 800VF	MicroTouch 800XF	MicroTouch 850XF
Density @ 77 °F (25 °C)	1.05	1.05	1.05	1.02
Maximum Particle Size (µm)	120	60	31	31
Mean Particle Size (d50, µm)	22-30	11-15	6-9	5-9

Examples of Formulated Tactile Effects

Pebbly Feel Using MicroTouch 800F

Formula:	Wt %:	Product:	Supplier:
AlberdingkUSA® U 2101	92.6%	60.5% solids PUD	Alberdingk-Boley
MicroTouch 800F	7.4%	polymeric additive	Micro Powders, Inc.
Total:	100.0%	(wet applied film thickness: 0.003"; 76µm)	

Fuzzy Feel Using MicroTouch 800VF

Formula:	Wt %:	Product:	Supplier:
Witcobond® W-505	92.6%	40.0% solids PUD	Chemtura
MicroTouch 800VF	7.4%	polymeric additive	Micro Powders, Inc.
Total:	100.0%	(wet applied film thickness: 0.003"; 76µm)	

Rubbery Feel Using MicroTouch 800XF

Formula:	Wt %:	Product:	Supplier:
AlberdingkUSA® U 2101	45.5%	60.5% solids PUD	Alberdingk-Boley
Bayhydrol® UH 240	45.5%	40.0% solids PUD	Bayer
MicroTouch 800XF	9.0%	polymeric additive	Micro Powders, Inc.
Total:	100.0%	(wet applied film thickness: 0.006"; 152µm)	

Silky Feel Using MicroTouch 850XF

Formula:	Wt %:	Product:	Supplier:
AlberdingkUSA® U 2101	26.0%	60.5% solids PUD	Alberdingk-Boley
Urotuf® L63-MPW-38	61.0%	38.0% solids PUD	Reichhold Inc.
MicroTouch 850XF	13.0%	polymeric additive	Micro Powders, Inc.
Total:	100.0%	(wet applied film thickness: 0.006"; 152µm)	

Gloss Reduction Data

	Pebbly		Fuzzy		Rubbery		Silky	
MT = MicroTouch	Control	w/MT	Control	w/MT	Control	w/MT	Control	w/MT
% additive*:	0	7.4	0	7.4	0	9	0	13
Gloss 20°	15	7	56	1	18	1	4	0.5
Gloss 60°	52	27	83	7	52	9	27	5
Gloss 85°	66	22	93	4	79	30	80	30

* based on total formula weight



About Micro Powders, Inc.

Micro Powders, Inc. is the worldwide leader in wax technology, manufacturing specialized micronized waxes, wax dispersions and emulsions for the paint, ink and coatings industries.

Micro Powders offers an extensive and innovative product range and is committed to providing consistent quality products, excellent customer service and superior technical support to its customers.

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